

HR DATA ANALYSIS PROJECT

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

#plot style
sns.set(style= 'whitegrid', palette= 'muted')
%matplotlib inline
```

```
In [2]: #loading the dataset
df = pd.read_excel(r"C:\Users\KRITIKA\Downloads\New HR.xlsx")
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Employee_Name	EmpID	Salary	PositionID	Position	State	Zip	DOB	Sex	M
0	Adinolfi, Wilson K	10026	62506	19	Production Technician I	MA	1960	1983-07-10	M	
1	Ait Sidi, Karthikeyan	10084	104437	27	Sr. DBA	MA	2148	1975-05-05	M	
2	Akinkuolie, Sarah	10196	64955	20	Production Technician II	MA	1810	1988-09-19	F	
3	Alagbe,Trina	10088	64991	19	Production Technician I	MA	1886	1988-09-27	F	
4	Anderson, Carol	10069	50825	19	Production Technician I	MA	2169	1989-09-08	F	

5 rows × 28 columns



```
In [4]: #Understand the Dataset Structure
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 311 entries, 0 to 310
Data columns (total 28 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Employee_Name                        311 non-null    object
1   EmpID                               311 non-null    int64
2   Salary                              311 non-null    int64
3   PositionID                          311 non-null    int64
4   Position                            311 non-null    object
5   State                               311 non-null    object
6   Zip                                  311 non-null    int64
7   DOB                                  311 non-null    datetime64[ns]
8   Sex                                  311 non-null    object
9   MaritalDesc                         311 non-null    object
10  CitizenDesc                         311 non-null    object
11  HispanicLatino                     311 non-null    object
12  RaceDesc                           311 non-null    object
13  DateofHire                         311 non-null    datetime64[ns]
14  DateofTermination                  104 non-null    datetime64[ns]
15  TermReason                         311 non-null    object
16  EmploymentStatus                   311 non-null    object
17  Department                         311 non-null    object
18  ManagerName                       311 non-null    object
19  ManagerID                         303 non-null    float64
20  RecruitmentSource                  311 non-null    object
21  PerformanceScore                   311 non-null    object
22  EngagementSurvey                   311 non-null    float64
23  EmpSatisfaction                    311 non-null    int64
24  SpecialProjectsCount               311 non-null    int64
25  LastPerformanceReview_Date         311 non-null    datetime64[ns]
26  DaysLateLast30                     311 non-null    int64
27  Absences                           311 non-null    int64
dtypes: datetime64[ns](4), float64(2), int64(8), object(14)
memory usage: 68.2+ KB

```

columns like DateofTermination and ManagerID has missing values

```

In [5]: #statistics
df.describe()

```

Out[5]:

	EmpID	Salary	PositionID	Zip	DOB
count	311.000000	311.000000	311.000000	311.000000	311
mean	10156.000000	69020.684887	16.845659	6555.482315	1979-02-06 09:48:02.315112544 22:50
min	10001.000000	45046.000000	1.000000	1013.000000	1951-01-02 00:00:00
25%	10078.500000	55501.500000	18.000000	1901.500000	1973-12-03 00:00:00
50%	10156.000000	62810.000000	19.000000	2132.000000	1980-09-30 00:00:00
75%	10233.500000	72036.000000	20.000000	2355.000000	1986-05-29 12:00:00
max	10311.000000	250000.000000	30.000000	98052.000000	1992-08-17 00:00:00
std	89.922189	25156.636930	6.223419	16908.396884	NaN



Salary ranges from ~30K to 110K Employees have up to 20 absences, with some having DaysLateLast30 = 0, indicating low tardiness Engagement Survey scores range between 1 and 5

In [6]: `#checking null values`
`df.isnull().sum()`

```
Out[6]: Employee_Name      0
EmpID                    0
Salary                  0
PositionID              0
Position                0
State                   0
Zip                     0
DOB                     0
Sex                     0
MaritalDesc             0
CitizenDesc             0
HispanicLatino          0
RaceDesc                0
DateofHire              0
DateofTermination       207
TermReason              0
EmploymentStatus        0
Department              0
ManagerName             0
ManagerID               8
RecruitmentSource        0
PerformanceScore         0
EngagementSurvey         0
EmpSatisfaction          0
SpecialProjectsCount     0
LastPerformanceReview_Date 0
DaysLateLast30          0
Absences                0
dtype: int64
```

DateofTermination: Missing in 207 records → likely still employed ManagerID: Missing in 8 records → may require exclusion

CLEANING THE DATASET

```
In [7]: df.columns = df.columns.str.strip().str.lower().str.replace(' ','_').str.replace(df.columns
```

```
Out[7]: Index(['employee_name', 'empid', 'salary', 'positionid', 'position', 'state',
              'zip', 'dob', 'sex', 'maritaldesc', 'citizendesc', 'hispaniclatino',
              'racedesc', 'dateofhire', 'dateoftermination', 'termreason',
              'employmentstatus', 'department', 'managername', 'managerid',
              'recruitmentsource', 'performancescore', 'engagementsurvey',
              'empsatisfaction', 'specialprojectscount', 'lastperformancereview_date',
              'dayslatelast30', 'absences'],
              dtype='object')
```

All column names are now in a clean, consistent format

CREATING DERIVED FEATURES

--Creating Column Age

```
In [8]: from datetime import datetime
current_year = datetime.now().year
df['Age'] = current_year - df['dob'].dt.year
```

--Creating Column tenure

```
In [9]: df['Tenure'] = (datetime.now() - df['dateofhire']).dt.days // 365
```

--Creating column isTerminated

```
In [10]: df['is_terminated'] = df['dateoftermination'].notnull().astype(int)
```

```
In [11]: df[['Age', 'Tenure', 'is_terminated']].head(10)
```

```
Out[11]:
```

	Age	Tenure	is_terminated
0	42	13	0
1	50	10	1
2	37	13	1
3	37	17	0
4	36	13	1
5	48	13	0
6	46	10	0
7	42	11	0
8	55	15	0
9	37	10	0

EXPLORATORY DATA ANALYSIS

--TOTAL EMPLOYEES

```
In [19]: employee_count = df['empid'].value_counts().sum()
employee_count
```

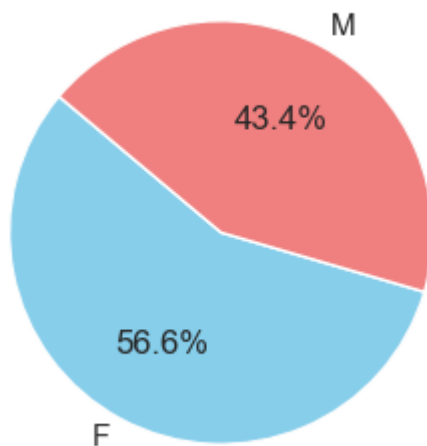
```
Out[19]: np.int64(311)
```

--Total employees 311

--TOTAL EMPLOYEES BY GENDER

```
In [16]: gender_counts = df['sex'].value_counts()
#ploting
plt.figure(figsize=(3,3))
plt.pie(gender_counts, labels=gender_counts.index, autopct='%1.1f%%', startangle=
plt.title('Employee Distribution by Gender')
plt.axis('equal')
plt.show()
```

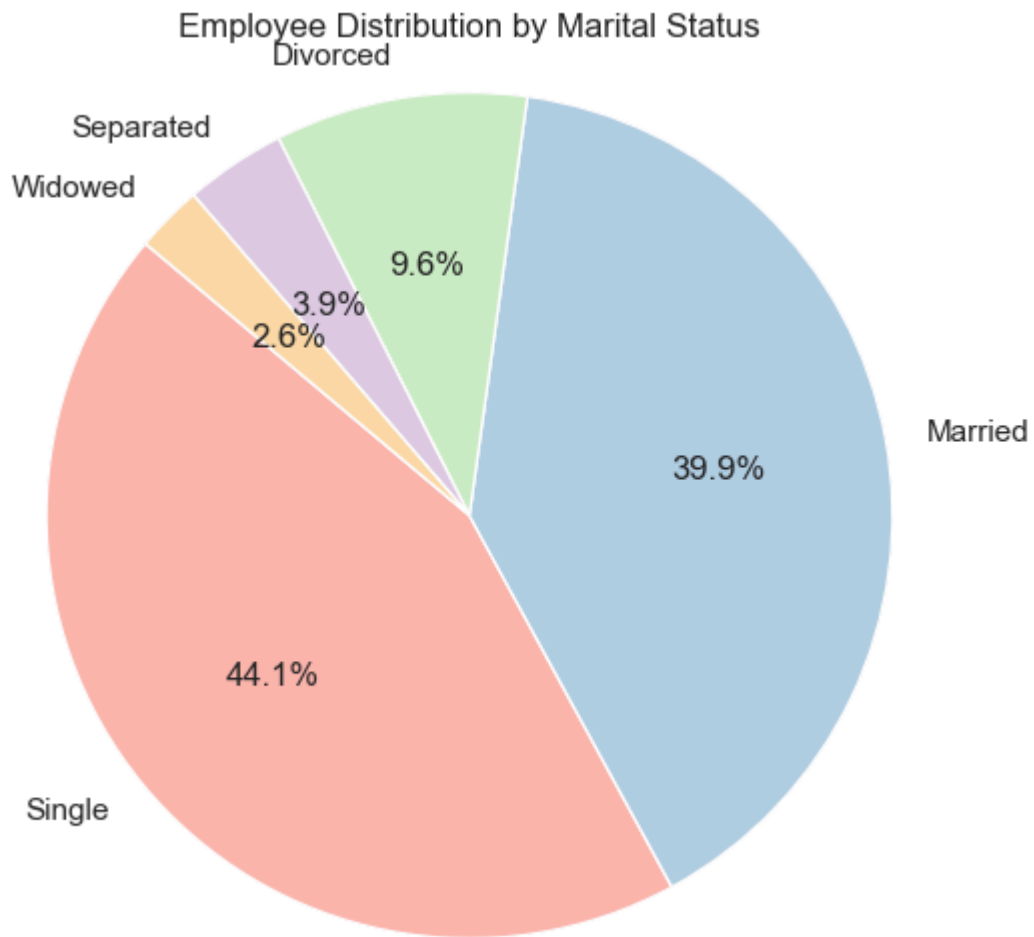
Employee Distribution by Gender



```
In [ ]: --Total female = 176
--Total male = 135
```

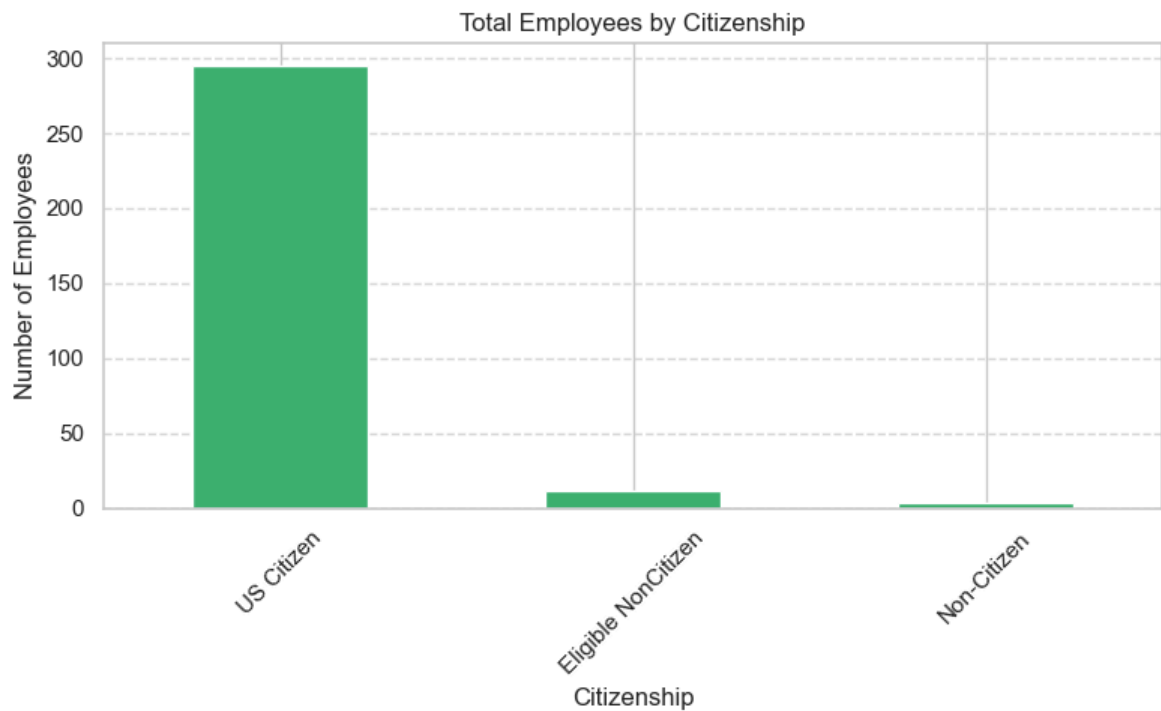
--TOTAL EMPLOYEES BY MARITAL STATUS

```
In [25]: marital_counts = df['maritaldesc'].value_counts()
plt.figure(figsize=(6,6))
plt.pie(marital_counts, labels=marital_counts.index, autopct='%1.1f%%', startangle=
plt.title('Employee Distribution by Marital Status')
plt.axis('equal')
plt.show()
```



-TOTAL EMPLOYEES BY CITIZENSHIP

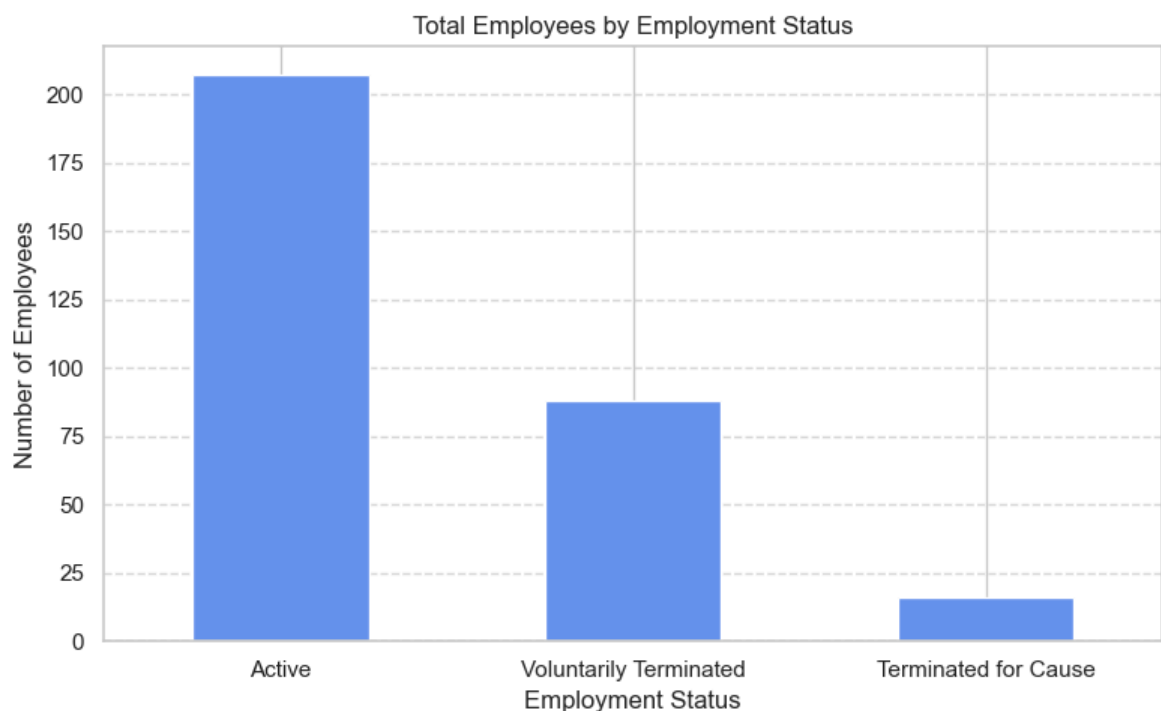
```
In [29]: citizenship_counts = df['citizendesc'].value_counts()
plt.figure(figsize=(8,5))
citizenship_counts.plot(kind='bar', color='mediumseagreen')
plt.title('Total Employees by Citizenship')
plt.xlabel('Citizenship')
plt.ylabel('Number of Employees')
plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



--A large number of employees are from US citizen.

-TOTAL NUMBER OF EMPLOYEES BY EMPLOYMENT STATUS

```
In [34]: employment_status_counts = df['employmentstatus'].value_counts()
plt.figure(figsize=(8,5))
employment_status_counts.plot(kind='bar', color='cornflowerblue')
plt.title('Total Employees by Employment Status')
plt.xlabel('Employment Status')
plt.ylabel('Number of Employees')
plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



--More than 200 employees are active. --Almost 75 - 90 employees are Voluntarily Terminated. --Less than 25 are terminated for cause

-TOTAL NUMBER OF EMPLOYEES BY MANAGER NAME

```
In [35]: manager_counts = df['managername'].value_counts()
plt.figure(figsize=(10,6))
manager_counts.plot(kind='bar', color='salmon')
plt.title('Total Employees by Manager')
plt.xlabel('Manager Name')
plt.ylabel('Number of Employees')
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



-SUM AND AVERAGE OF SALARY

```
In [36]: total_salary = df['salary'].sum()
average_salary = df['salary'].mean()

print(f"Total Salary Paid to All Employees: ₹{total_salary:,.2f}")
print(f"Average Salary of Employees: ₹{average_salary:,.2f}")
```

Total Salary Paid to All Employees: ₹21,465,433.00

Average Salary of Employees: ₹69,020.68

--OVERALL TERMINATION STATUS

```
In [12]: termination_count = df['is_terminated'].value_counts()
termination_count
```

```
Out[12]: is_terminated
0      207
1      104
Name: count, dtype: int64
```

```
In [16]: #ploting
plt.figure(figsize=(7,5))
ax = sns.barplot(x = termination_count.index, y = termination_count.values, pale
```



```

for bar in ax.patches:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 1,
        f'{int(height)}',
        ha = 'center', va = 'bottom', fontsize=12, fontweight='bold'
    )

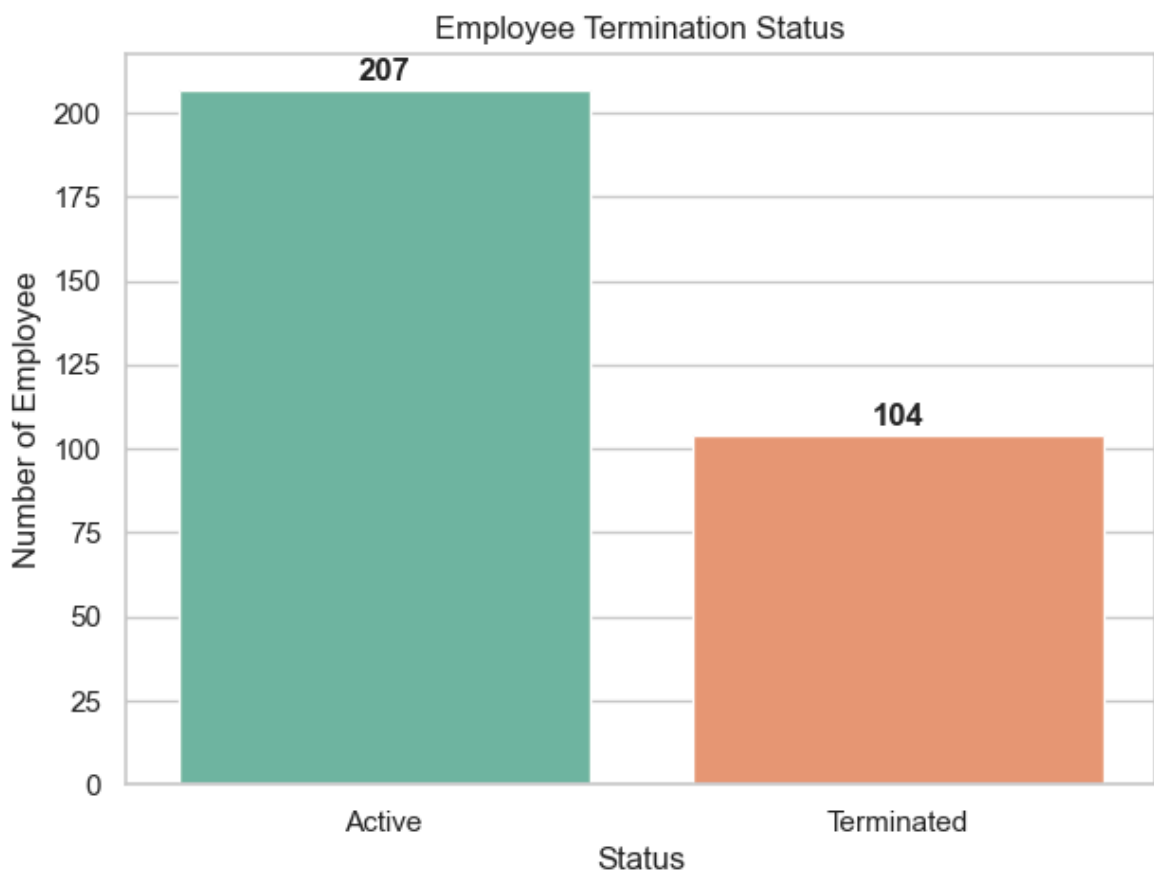
plt.xticks([0,1],['Active','Terminated'])
plt.title('Employee Termination Status')
plt.xlabel('Status')
plt.ylabel('Number of Employee')
plt.show()

```

C:\Users\KRITIKA\AppData\Local\Temp\ipykernel_14144\3128851600.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
ax = sns.barplot(x = termination_count.index, y = termination_count.values, palette = 'Set2')
```



--Active Employees - 207 --Terminated Employees - 104

-TERMINATION BY DEPARTMENT

```

In [17]: terminated = df[df['is_terminated'] ==1 ]
         dept_count = terminated['department'].value_counts()

         #ploting

```

```
plt.figure(figsize=(10,6))
ax = sns.barplot(x= dept_count.index, y= dept_count.values, palette = 'pastel')
for bar in ax.patches:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 1,
        f'{int(height)}',
        ha='center', va='bottom', fontsize=11, fontweight='bold'
    )

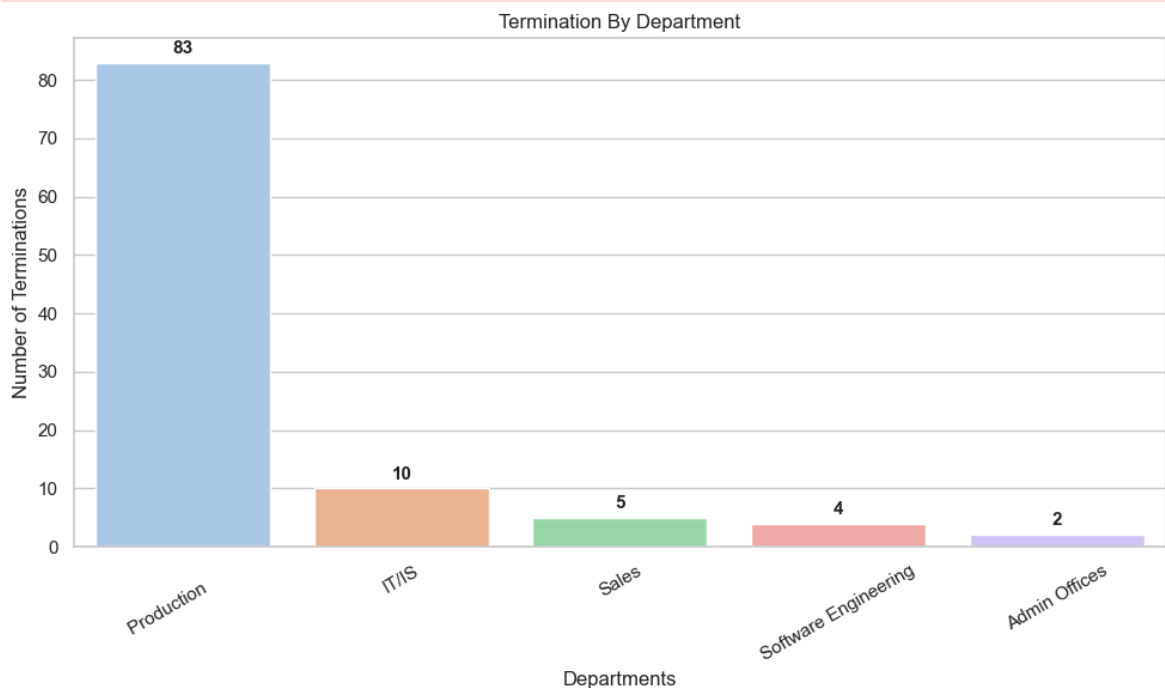
plt.title('Termination By Department')
plt.xlabel('Departments')
plt.ylabel('Number of Terminations')
plt.xticks(rotation=30)
plt.tight_layout()

plt.show()
```

C:\Users\KRITIKA\AppData\Local\Temp\ipykernel_14144\3381452627.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
ax = sns.barplot(x= dept_count.index, y= dept_count.values, palette = 'pastel')
```



-Production department has the highest number of terminated employees.

--TERMINATION DISTRIBUTION BY GENDER

```
In [18]: terminated = df[df['is_terminated'] == 1]
gender_count = terminated['sex'].value_counts()
labels = gender_count.index
colors = ['#66c2a5', '#fc8d62']

#ploting
plt.figure(figsize=(3,3))
plt.pie(
```

```

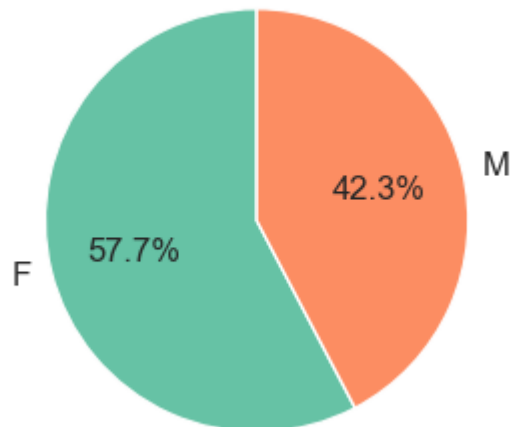
gender_count,
labels=labels,
autopct= '%1.1f%%',
startangle = 90,
colors = colors,
textprops={'fontsize' : 12})

plt.title('Gender Distribution Among Terminated Employees')
plt.axis('equal')

plt.show()

```

Gender Distribution Among Terminated Employees



- Total Terminated Employees: 104 - Males: 44 (42.3%) - Females: 60 (57.7%)

--AGE DISTRIBUTION OF TERMINATED EMPLOYEE

```

In [19]: terminated = df[df['is_terminated'] == 1]

#plot
plt.figure(figsize=(6,4))
sns.boxplot(y= terminated['Age'], color='#8da0cb')

plt.title('Age Distribution Of Terminated Employees')
plt.ylabel('Age')
plt.grid(axis='y', linestyle='--', alpha= 0.5)
plt.show

```

Out[19]: <function matplotlib.pyplot.show(close=None, block=None)>



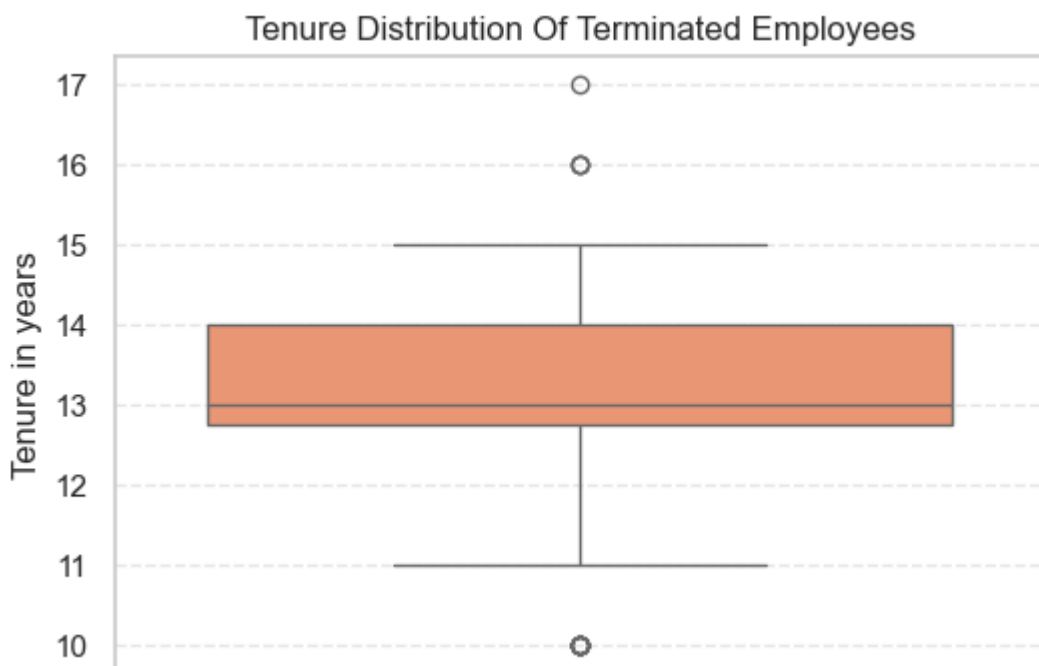
-Median age of terminated employees is around **45-47 years** - Most terminations occur between **45 to 55 years** - A few older employees also got terminated (visible as outliers)

--TENURE DISTRIBUTION OF TERMINATED EMPLOYEES

```
In [20]: terminated = df[df['is_terminated'] == 1]

#plot
plt.figure(figsize=(6,4))
sns.boxplot(y= terminated['Tenure'], color='#fc8d62')

plt.title('Tenure Distribution Of Terminated Employees')
plt.ylabel('Tenure in years')
plt.grid(axis = 'y', linestyle= '--', alpha = 0.5)
plt.show()
```



-Median tenure of terminated employees is approximately **13 - 13.5 years** - Most terminations occur between **12 and 15 years** of service - Some employees were terminated after even **16+ years**, indicating long-term

separations and some terminated within 10 years indicating short-term separation

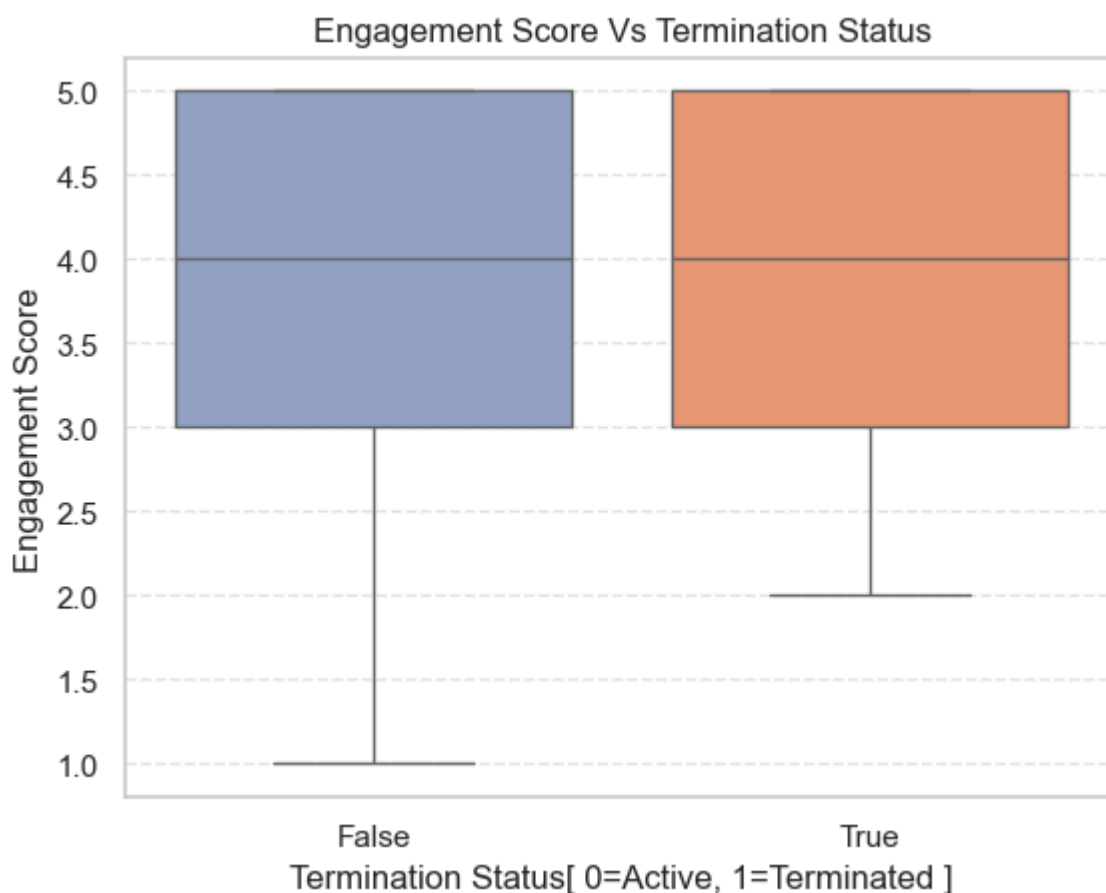
--EMPLOYEE ENGAGEMENT SCORE VS TERMINATION STATUS

```
In [66]: sns.boxplot(x= 'is_terminated' , y= 'empsatisfaction', data=df, palette=['#8da0cb', '#fc8d62'])
plt.title('Engagement Score Vs Termination Status')
plt.xlabel('Termination Status[ 0=Active, 1=Terminated ]')
plt.ylabel('Engagement Score')
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.show()
```

C:\Users\KRITIKA\AppData\Local\Temp\ipykernel_14144\4052678806.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x= 'is_terminated' , y= 'empsatisfaction', data=df, palette=['#8da0cb', '#fc8d62'])
```



-Active employees shows a higher median engagement score and a wider range of score compared to Terminated employees.

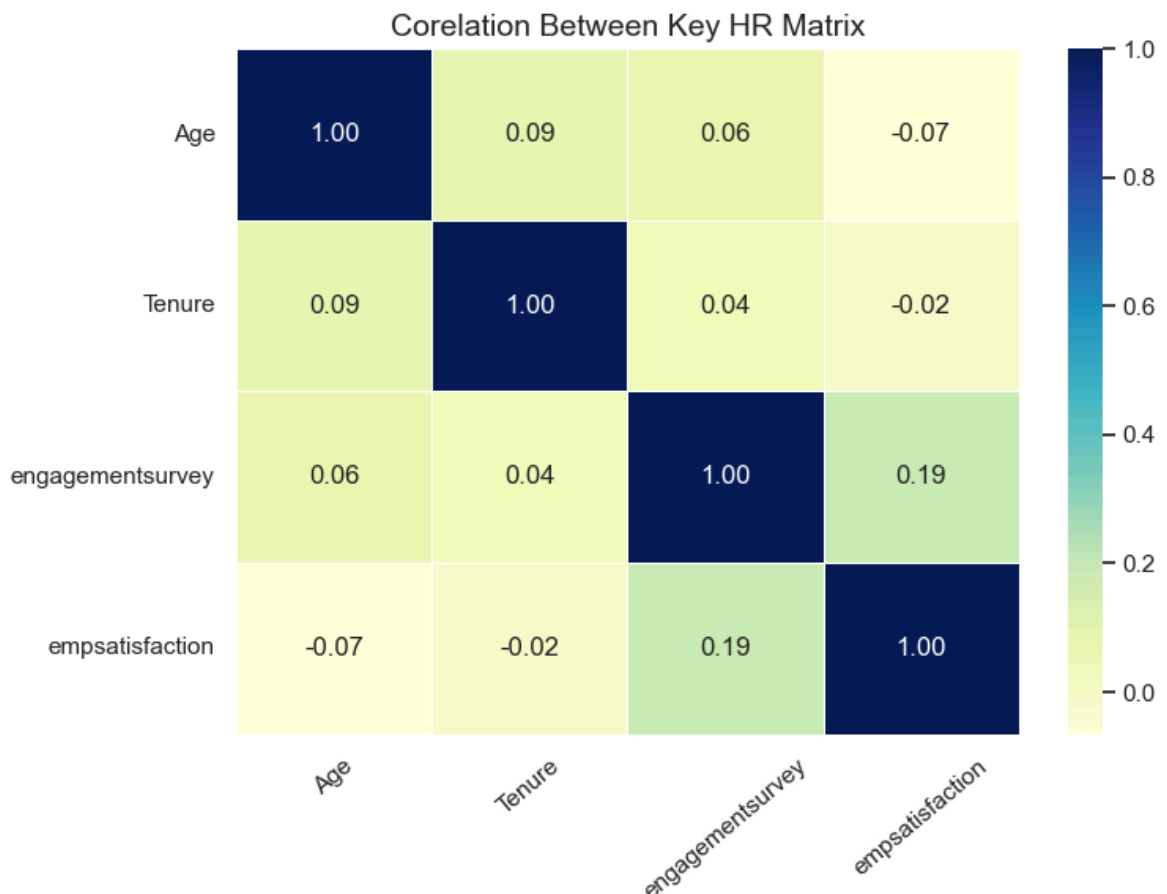
-CORRELATION HEATMAP FOR KEY NUMERIC VARIABLES

```
In [28]: corr_features = df[['Age', 'Tenure', 'engagementsurvey', 'empsatisfaction']]
corr_matrix = corr_features.corr()

plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix, annot=True, fmt= '.2f', cmap='YlGnBu', linewidths=0.5)

plt.title('Corelation Between Key HR Matrix', fontsize=14)
plt.xticks(rotation=40)
```

```
plt.tight_layout()
plt.show()
```



--There is strong correlation of 0.19 between employee engagement and employee satisfaction. --There is a weak positive correlation of 0.09 between age and tenure. --Negative correlation between age and tenure shows minimal impact on employee satisfaction.

-EMPLOYEE SATISFACTION BY DEPARTMENT

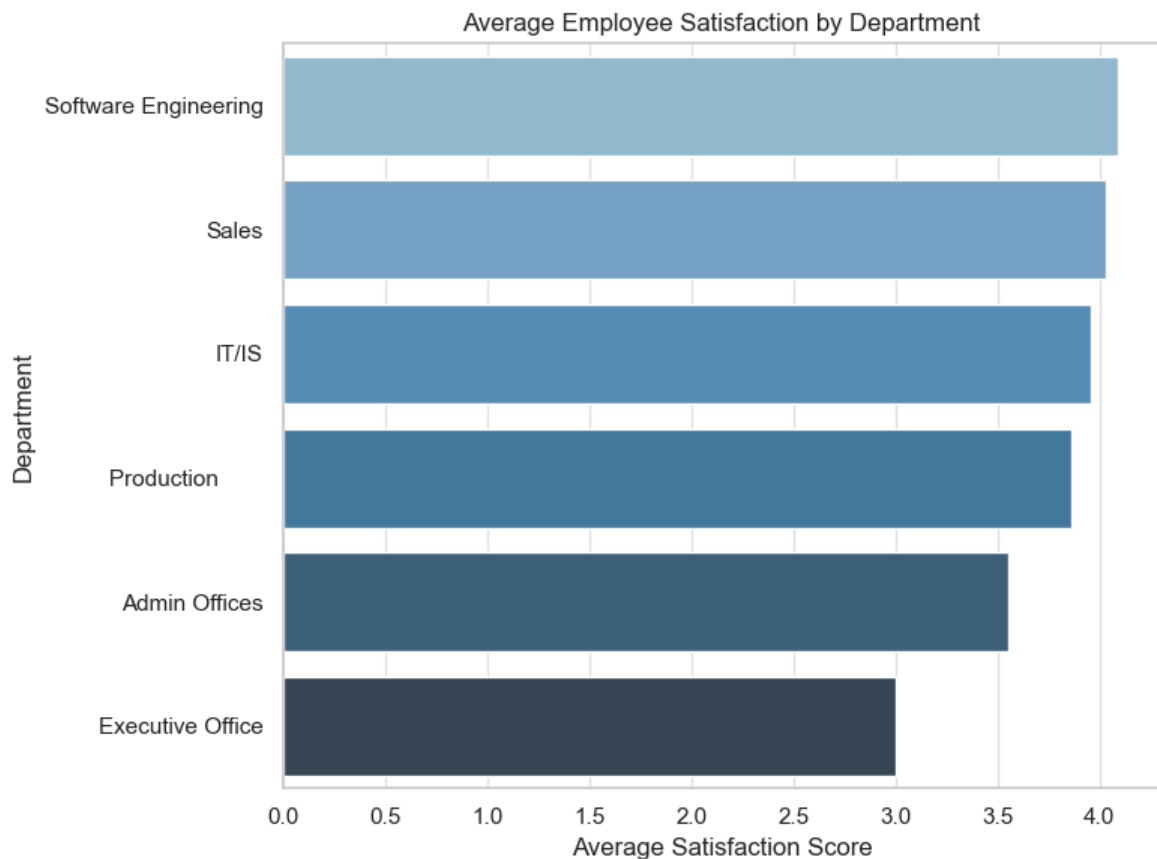
```
In [33]: dept_satisfaction = df.groupby('department')['empsatisfaction'].mean().sort_values
plt.figure(figsize=(8,6))
sns.barplot(y= dept_satisfaction.index, x= dept_satisfaction.values, palette='Blues')

plt.title('Average Employee Satisfaction by Department')
plt.xlabel('Average Satisfaction Score')
plt.ylabel('Department')
plt.grid(axis='x', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

C:\Users\KRITIKA\AppData\Local\Temp\ipykernel_14144\4113021662.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(y= dept_satisfaction.index, x= dept_satisfaction.values, palette='Blues_d')
```



--Executive Office department have the lowest average satisfaction score while the other department have higher scores ranging from 3.5 to more than 4.0.

-ENGAGEMENT SCORE BY DEPARTMENT

```
In [37]: dept_engagement = df.groupby('department')['engagementsurvey'].mean().sort_values

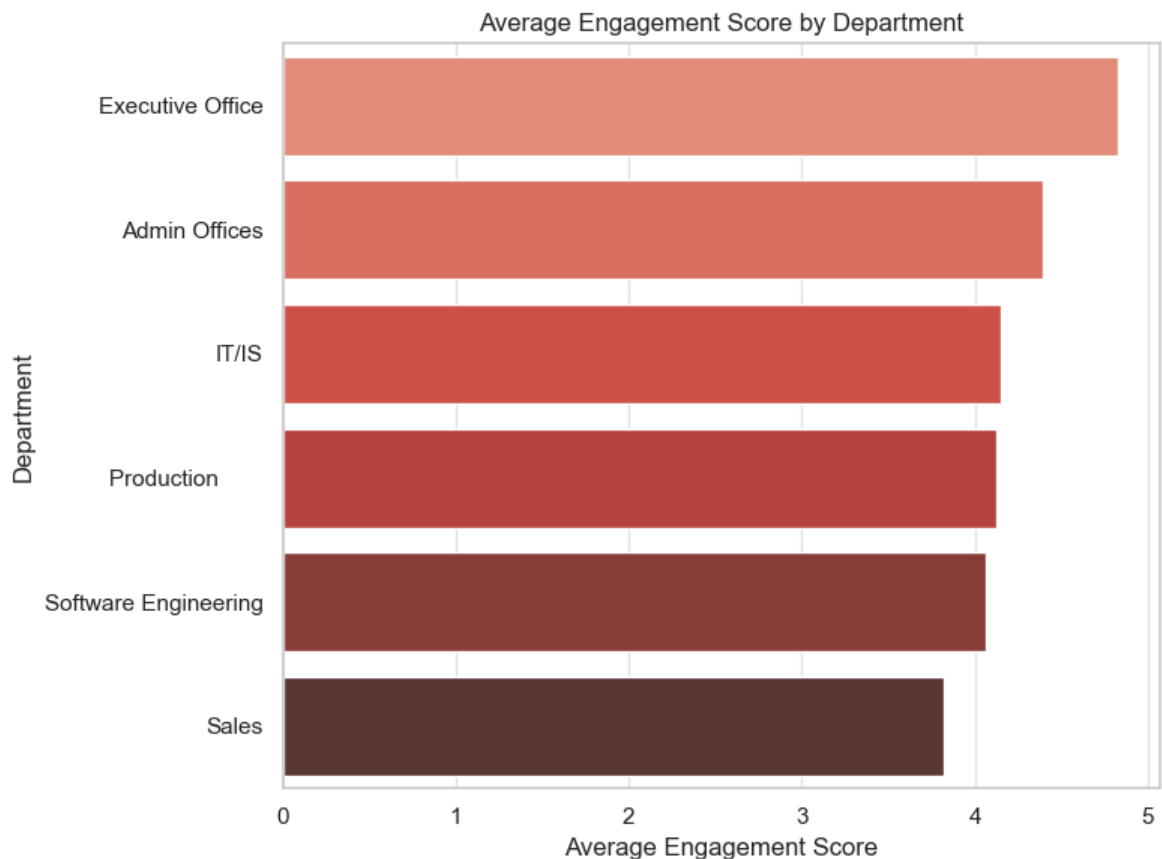
plt.figure(figsize=(8,6))
sns.barplot(y= dept_engagement.index, x= dept_engagement.values, palette='Reds_d

plt.title('Average Engagement Score by Department')
plt.xlabel('Average Engagement Score')
plt.ylabel('Department')
plt.grid(axis='x', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

C:\Users\KRITIKA\AppData\Local\Temp\ipykernel_14144\3055311766.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(y= dept_engagement.index, x= dept_engagement.values, palette='Reds_d')
```



- Departments like **Executive and Admin offices** have the **highest engagement** - Department such as **Sales** show lower engagement scores

-TERMINATION BY PERFORMANCE SCORE

```
In [63]: terminated_df = df[df['dateoftermination'].notna()]

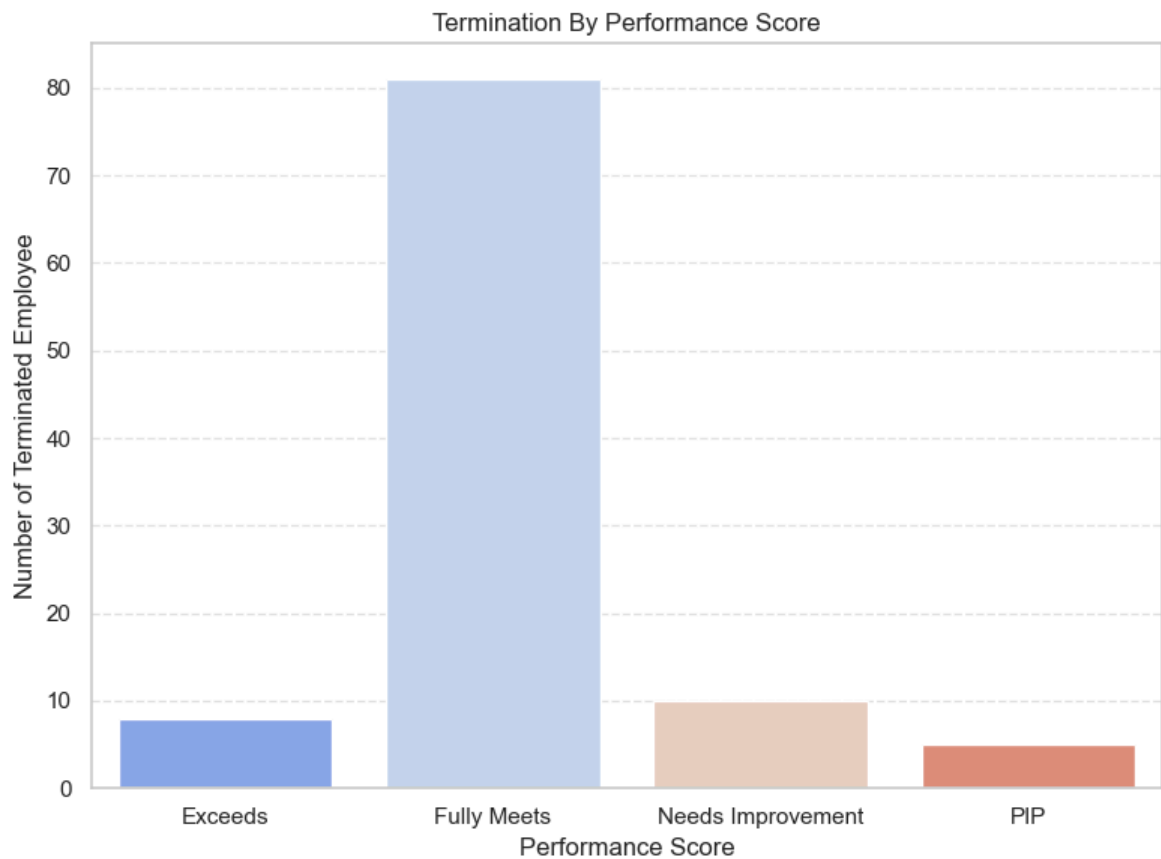
perf_termination = terminated_df['performancescore'].value_counts().reindex(df['performancescore'].value_counts().index)

plt.figure(figsize=(8,6))
sns.barplot(x=perf_termination.index , y= perf_termination.values, palette= 'coolwarm')
plt.title('Termination By Performance Score')
plt.xlabel('Performance Score')
plt.ylabel('Number of Terminated Employee')
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

C:\Users\KRITIKA\AppData\Local\Temp\ipykernel_14144\3651234695.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=perf_termination.index , y= perf_termination.values, palette= 'coolwarm', legend= False)
```

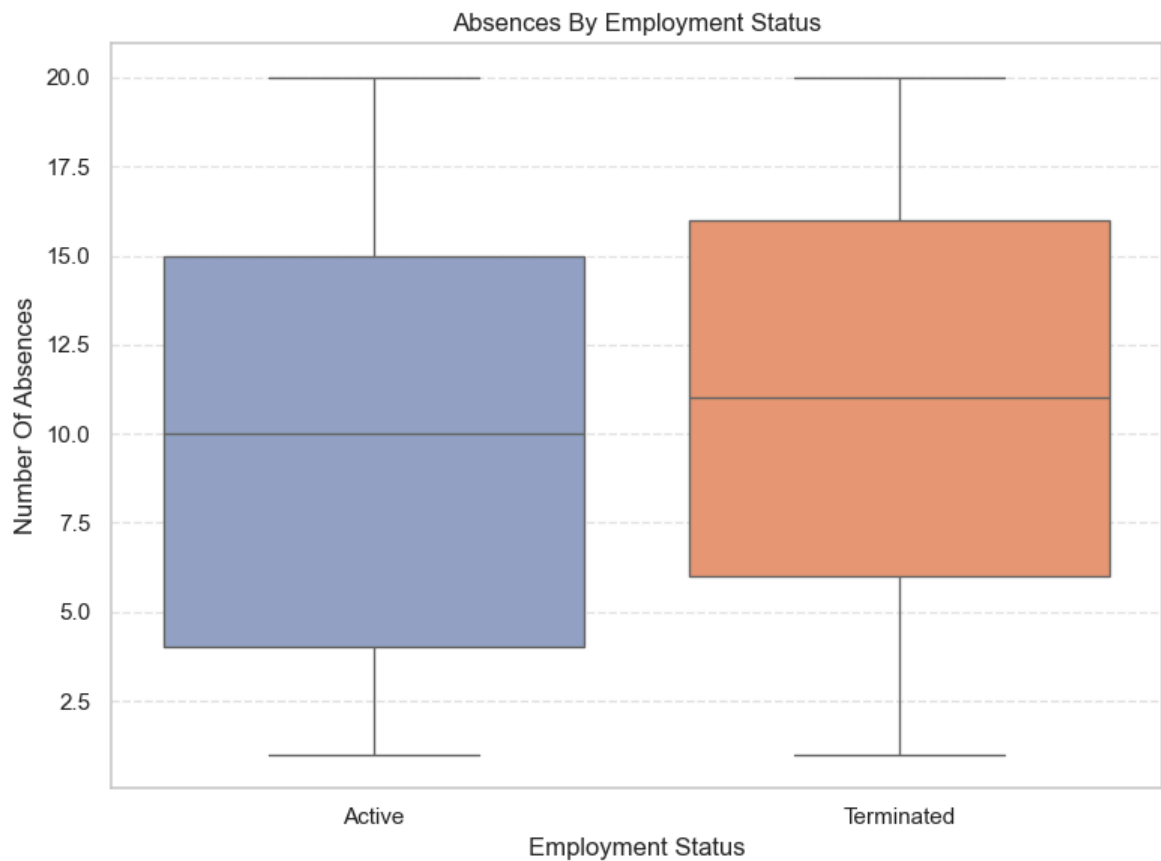



- Most terminations occurred among employees rated as "Fully Meets" followed by "Needs Improvement" - Very few terminations from "Exceeds Expectations" or "PIP" categories

-ABSENCES BY TERMINATION STATUS

```
In [58]: df['is_terminated'] = df['dateoftermination'].notna()

plt.figure(figsize=(8,6))
sns.boxplot(x= 'is_terminated' , y='absences' ,data=df, palette = ['#8da0cb', '#f
plt.xticks([0,1],['Active','Terminated'])
plt.title('Absences By Employment Status')
plt.xlabel('Employment Status')
plt.ylabel('Number Of Absences')
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```



--There is a slight difference between the median of the active and terminated number of absences.

-TENURE VS TERMINATION STATUS

```
In [57]: plt.figure(figsize=(8,6))
sns.violinplot(x= 'is_terminated', y= 'Tenure', data=df, palette=['#8da0cb', '#f
plt.xticks([0,1],['Active', 'Terminated'])
plt.title('Tenure Distribution By Employment Status')
plt.xlabel('Employment Status')
plt.ylabel('Tenure in Years')
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```



--Large number of termination occurs after ****12-14 years**** --Active employees have a slightly wider distribution extended upto 20 years.

This project focuses on analyzing employee data to uncover insights related to:

- Termination trends
- Employee engagement and satisfaction
- Absenteeism patterns
- Tenure and performance correlations

Key Insights

1. **Total Employees** -Total female = 176 -Total male = 135 -Most of them are Single - A large number of employees are from US citizen.
2. **Employment Status** -More than 200 employees are active. -Almost 75 - 90 employees are Voluntarily Terminated. -Less than 25 are terminated for cause
3. **Termination Trends**
 - Highest termination rate observed in the *Production* department.
 - More female employees were terminated compared to male employees.
 - Most terminations occur between **45 to 55 years**
4. **Salary details**
 - Total Salary Paid to All Employees: ₹21,465,433.00 -Average Salary of Employees: ₹69,020.68
5. **Performance & Satisfaction**

- Employees with *Low* performance scores had significantly higher termination rates.
- Terminated employees showed lower satisfaction and engagement scores on average.
- strong corelation of 0.19 between employee engagement and employee satisfaction

6. **Employee Satisfaction & Department** -Executive Office department have the lowest average satisfaction score while the other department have higher scores ranging from 3.5 to more than 4.0.

- Departments like **Executive and Admin offices** have the **highest engagement**
- Department such as **Sales** show lower engagement scores

7. **Absenteeism**

- Terminated employees had a slightly higher range of absences.
- High absenteeism can be a potential early indicator of disengagement.

8. **Tenure Analysis** -Most terminations occur between **12 and 15 years** of service

- Some employees were terminated after even **16+ years**
- Active employees generally had longer tenures and more stable performance scores.
- There is a weak positive correlation of 0.09 between age and tenure shows minimal impact on employee satisfaction..

Tools Used

- **Python Libraries:** Pandas, Matplotlib, Seaborn
 - **Jupyter Notebook:** For data exploration and visualization
 - **Next Step:** This analysis will be replicated in **Power BI** for interactive dashboarding.
-

By: Kritika Naidu

In []: